



Group 110

Toni Rämö

34

▲ 5

Group 110



main

0.6889

3

▼ 1

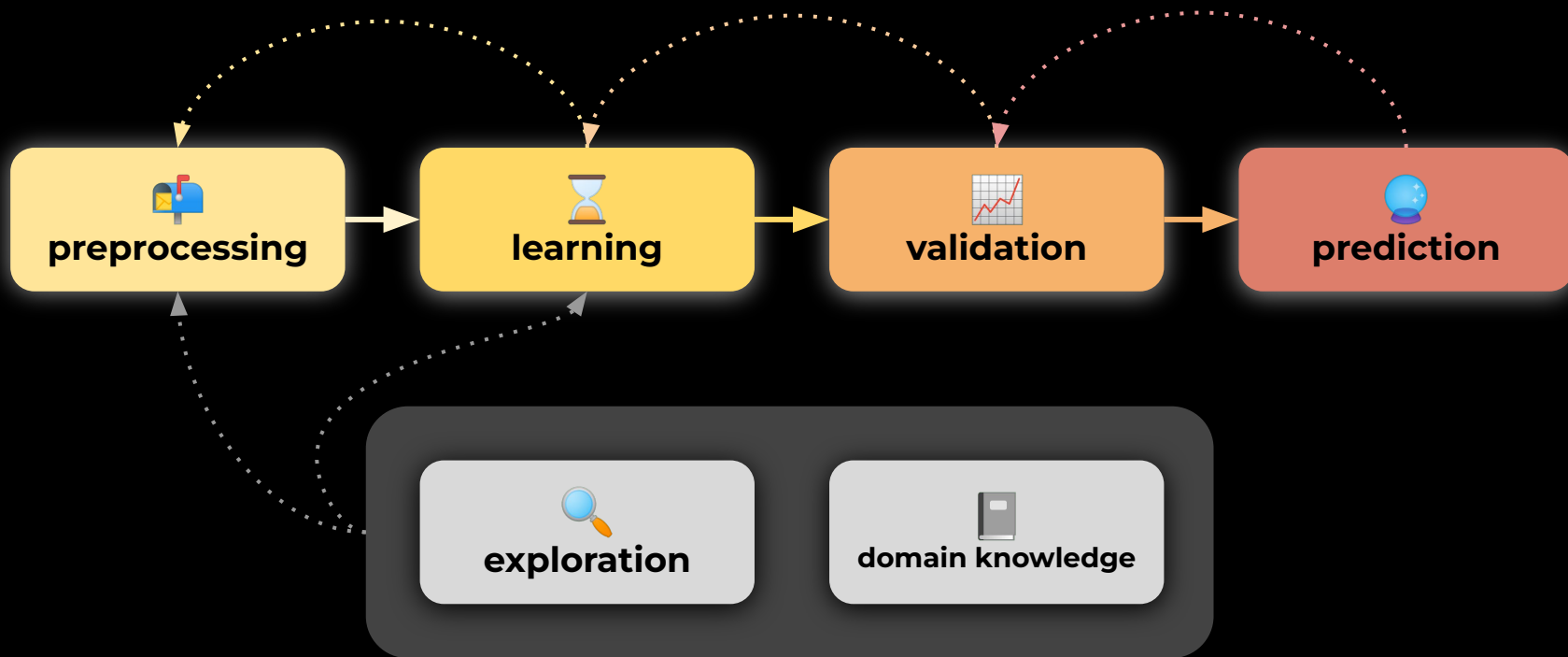
Group 110



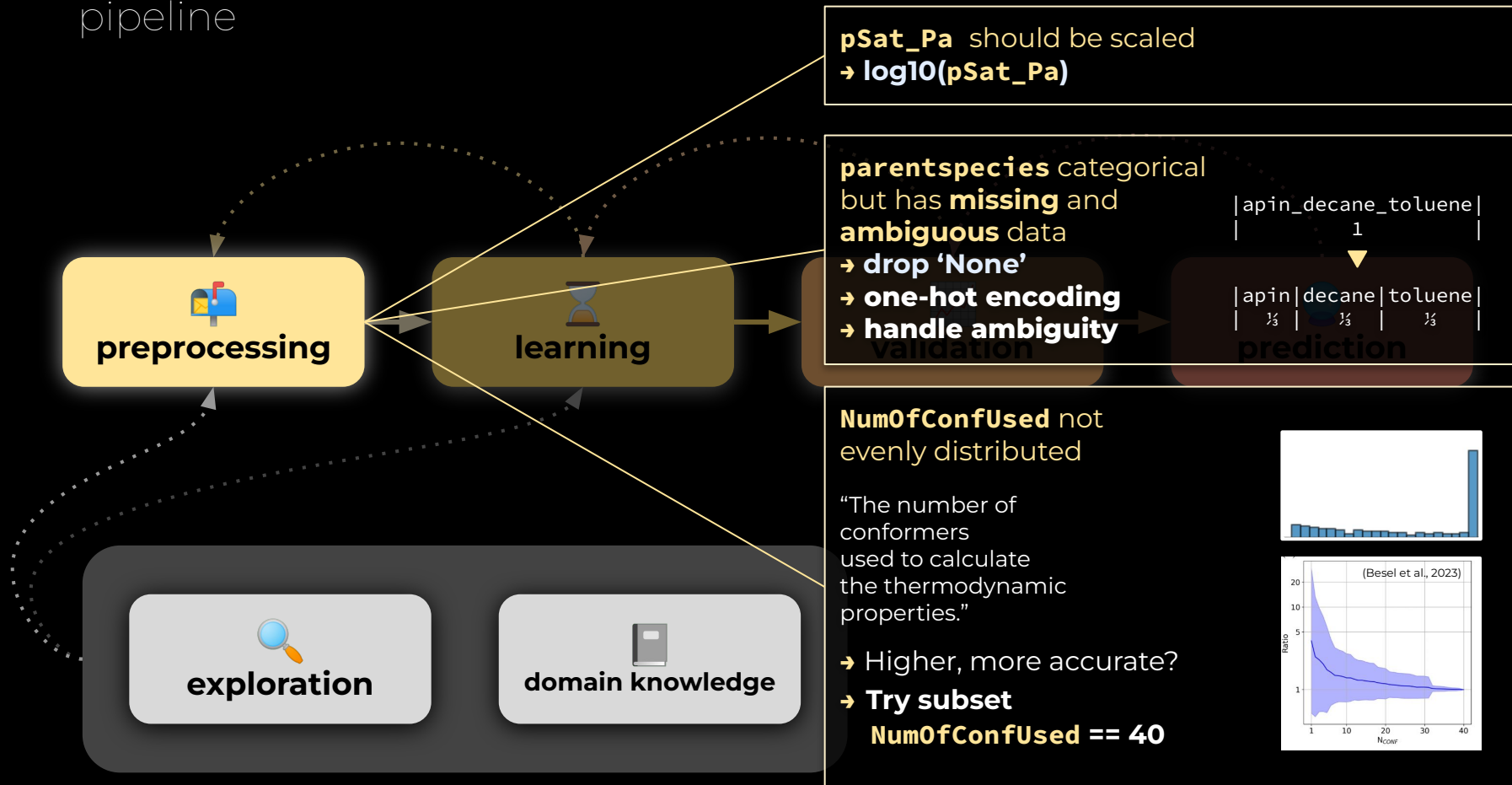
ADV

0.77129

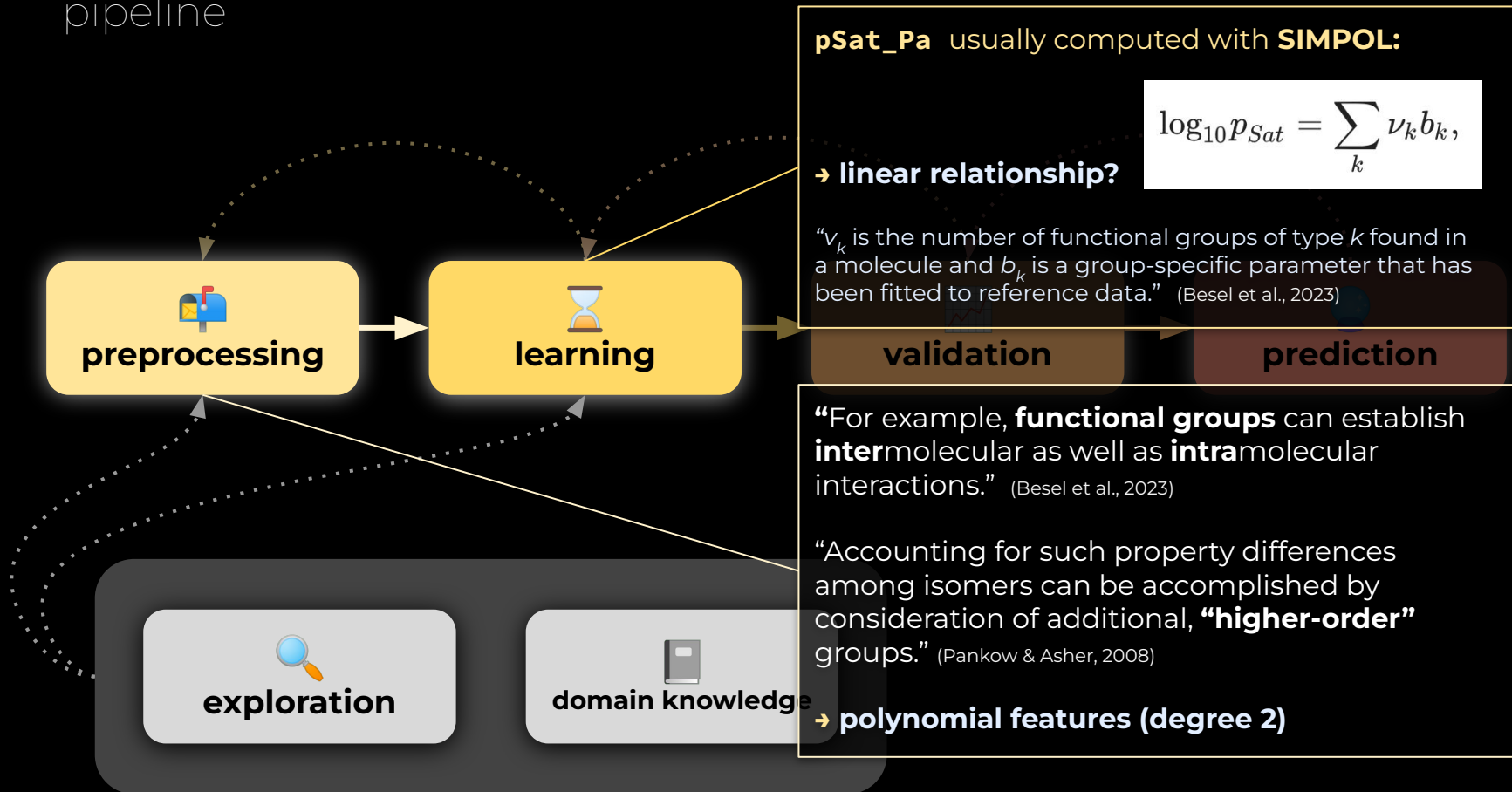
pipeline



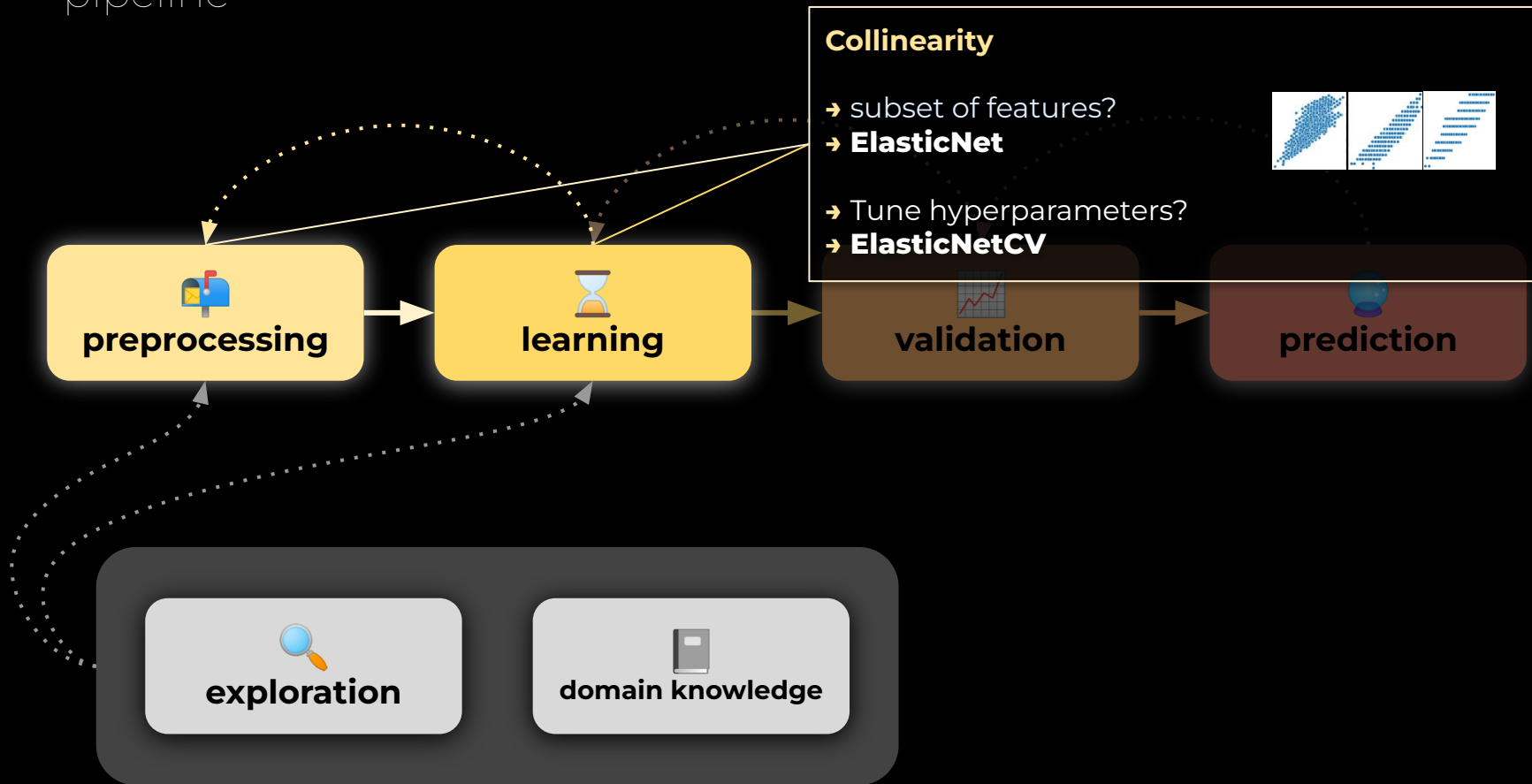
pipeline



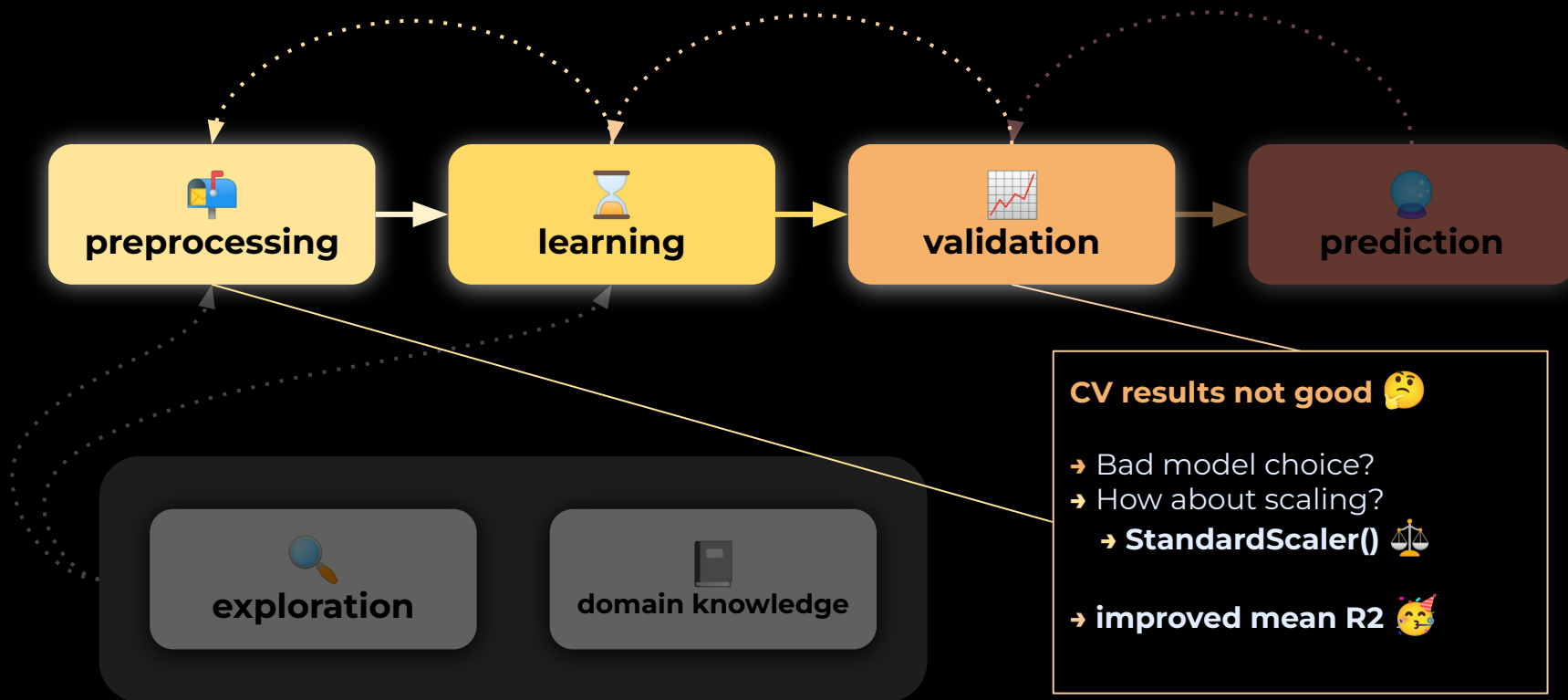
pipeline



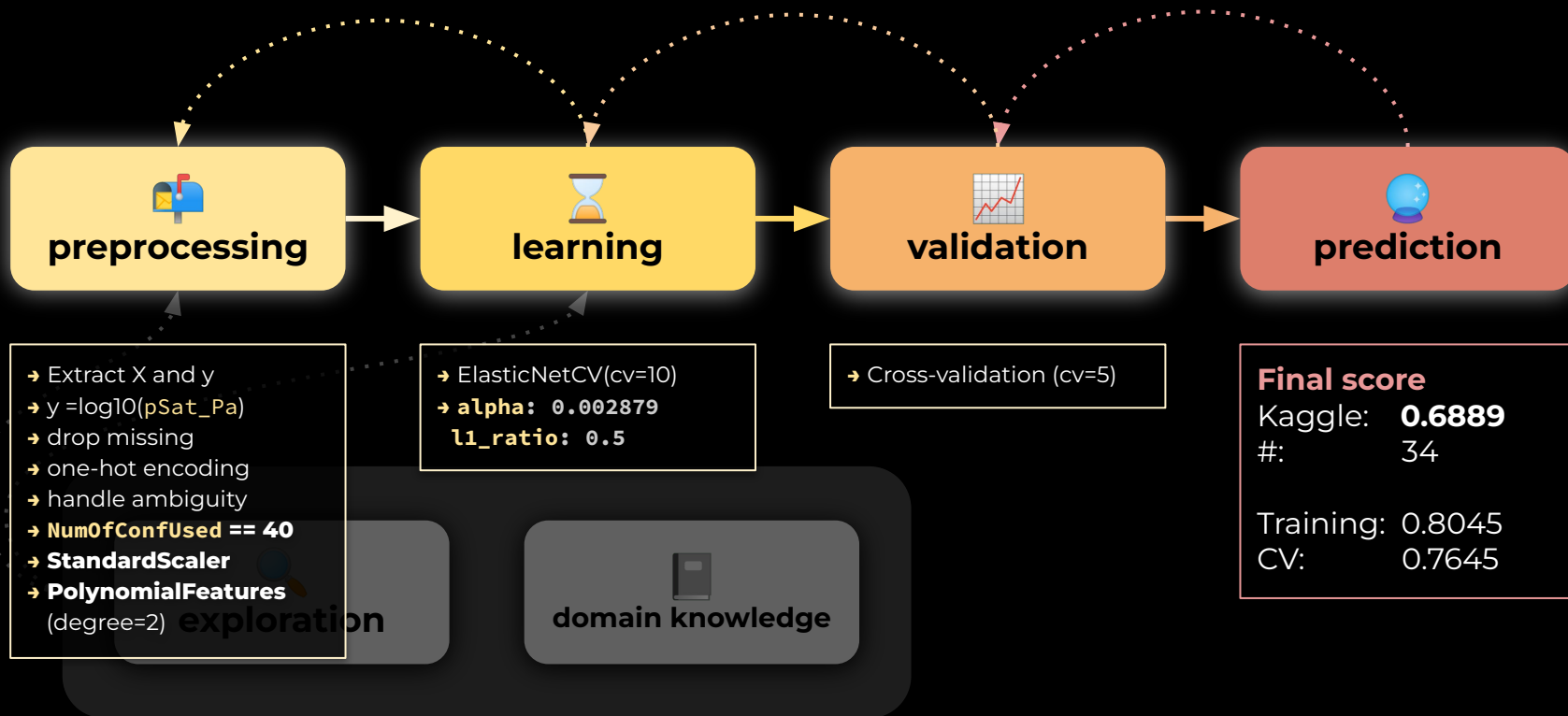
pipeline



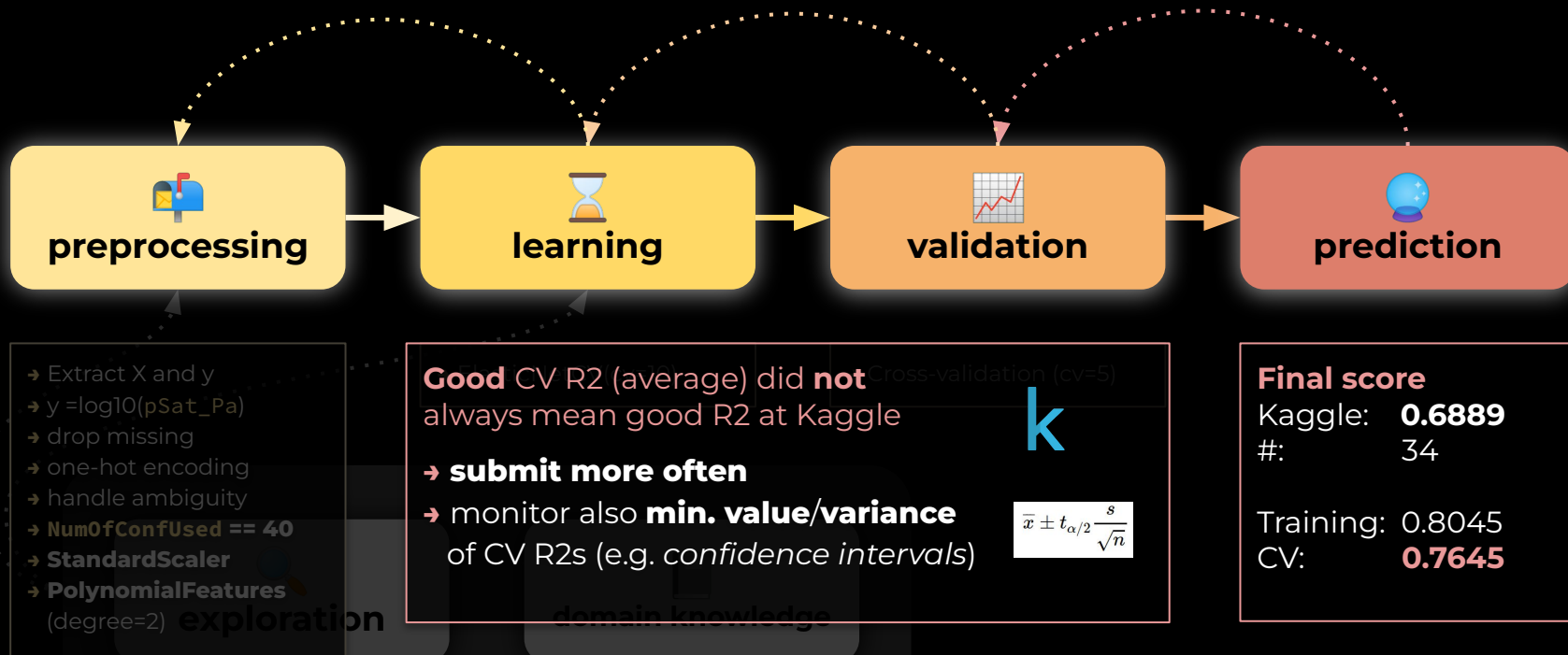
pipeline



results



some learnings & observations





It is better **not to change too many parameters** at the same time

However, **the number of different combinations explodes** easily.

- How to be efficient?
- How to choose a right path?...



...Helper functions, implemented early on, can make experimenting at least more convenient.

- `sklearn.pipeline.make_pipeline(*steps)`
- `train_and_validate(pipe)`
- `train_and_get_test_targets(pipe)`



preprocessing



learning



validation



prediction

- Extract X and y
- `y = log10(pSat_Pa)`
- drop missing
- one-hot encoding
- handle ambiguity
- `NumOfConfUsed == 40`
- `StandardScaler`
- `PolynomialFeatures`
(degree=2)

Good CV R2 (average) did not always mean good R2 at Kaggle

- **submit more often**
- monitor also **min. value/variance** of CV R2s (e.g. *confidence intervals*)

k

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

Final score

Kaggle: **0.6889**
#: 34

Training: 0.8045
CV: **0.7645**

some learnings & observations

Solution may not be the best, but at least the model is

→ Interpretable 🤖

403 coeff.s
in total
from which
240 are 0
18 are ≥ 0.1

- Extract X and y
- $y = \log_{10}(p_{\text{Sat}}/Pa)$
- top 20 coeff.s →
- one-hot encoding
- handle ambiguity
- NumOfConfUsed == 40
- StandardScaler
- PolynomialFeatures (degree=2)

feature	coefficient
intercept	-3.781213
NumOfC	-0.942855
NumHBondDonors	-0.793636
NumOfConf	-0.504708
carboxylic.acid	-0.306933
carbonylperoxynitrate	0.285672
NumOfAtoms NumOfO	-0.217580
parentspecies_decane^2	-0.187735
NumOfO parentspecies_decane	-0.169739
peroxide parentspecies_toluene	-0.165673
NumOfO ketone	0.130561

feature	coefficient
aldehyde ketone	0.124722
NumOfAtoms^2	0.120224
NumOfC ester	0.117123
aldehyde	-0.113261
parentspecies_decane	0.113020
parentspecies_toluene	
hydroperoxide	-0.110660
NumOfAtoms	0.110251
C.C..non.aromatic. ketone	-0.099262
carbonylperoxyacid	-0.087952
parentspecies_toluene	

→ Fast to train and predict ▶▶

results

